

WILLIAM LAZCANO

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EDUCATION

B.S. Computer & Electrical Engineering — Dual Major

2022 – 2028 (Expected)

University of Nevada, Las Vegas | Coursework: Circuit Analysis, EE 220 Circuits, Digital Logic, Systems Programming, C++, Data Structures

TECHNICAL SKILLS

Electrical: Circuit analysis, KVL/KCL, Thevenin equivalents, RC circuits, load calculations, NEC-compliant wiring

Simulation: LTspice (SPICE netlists), ModelSim (Verilog HDL), C++ numerical simulation

HDL: Verilog, SystemVerilog

Programming: C++, Python

Tools: LTspice, ModelSim, Quartus II, Git, Linux/Unix

Languages: Spanish (Native), English (Fluent)

ENGINEERING PROJECTS

Battery Discharge Simulator — Multi-Platform

Fall 2025

- Modeled 3S2P 18650 Li-ion pack (11.1V, 6000 mAh) across three platforms: C++ OOP simulation, LTspice SPICE netlist, and Verilog HDL in ModelSim
- C++ used configurable BatterySimulator class; LTspice modeled Thevenin equivalent with internal resistance and SOC-dependent voltage
- Verilog module used clock-driven state updates with SOC register and battery_depleted output flag
- All platforms converged within 1% variance: 5.8h runtime, 10.3 Wh energy — validated against hand calculations using KVL and Ohm's Law

GCD Hardware Implementation

Fall 2024

- Implemented Euclidean GCD in hardware using 6-state FSM and Verilog datapath on Altera DE2-115 FPGA
- 122 MHz max frequency, +11.8 ns slack, 87 logic elements (0.08%); verified across 1,000 random test cases

Custom 8-Bit Multicycle Processor

Fall 2024

- Designed complete 8-bit CPU from scratch in SystemVerilog — 16-instruction ISA, FSM control unit, deployed on FPGA at 50 MHz

PROFESSIONAL EXPERIENCE

Construction & Electrical Specialist | Lazcano Family Construction

2019 – Present

- NEC-compliant residential electrical: 200A panel upgrades, knob-and-tube removal, GFCI/AFCI across 50+ projects; 100% inspection pass rate
- Apply circuit analysis and load calculations daily — wire sizing, breaker selection, 80% derating, ground resistance verification with megger
- 3 full home rewires, 8 bathroom renovations, 8,000+ sq ft flooring — systematic fault tracing using tone generators and multimeters

Team Member | Target Corporation

May 2023 – Present

- High-volume retail customer service; bilingual Spanish/English

Lifeguard | MGM Resorts International

2020 – 2022

- Monitored guest safety at high-traffic resort; maintained certifications, responded to emergencies

Portfolio & project documentation: will-l10.github.io